

Daniel L. Perret, PhD

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OVERVIEW

I am a **forest ecologist and conservation biogeographer** working to untie the stubborn ecological knots at the center of applied forest conservation and management challenges. My recent research foci include **demographic pathways of forest resilience to fire and climate change**, spatiotemporal predictability in **ecological forecasting**, and integration of field, aerial, and satellite data for **estimating biological forest disturbances**. I use many different tools and sources of information in my work, including forest inventories, tree-ring time series, Bayesian hierarchical models, aerial and satellite remote sensing data, future climate projections, and cutting-edge ecological theory.

EDUCATION

- PhD, Brown University** 2022
Niche-based perspectives on species' responses to climate change: pines as a multi-scale case study
Department of Ecology, Evolutionary & Organismal Biology
Advisor: Prof. Dov F. Sax
- BS, Stanford University** 2013
Biology (Ecology and Evolutionary Biology, *with Honors*)
Advisor: Prof. Elizabeth A. Hadly

PROFESSIONAL EXPERIENCE

- Postdoctoral Research Fellow** 2022 - present
United States Forest Service Pacific Northwest Research Station
Oak Ridge Institute for Science and Education
- Lead researcher on projects evaluating temperate forest resilience to climate and disturbance using forest inventories, tree-ring data, field surveys, remote sensing data, and Bayesian hierarchical models
 - Publications garnered invited commentaries by international scientists in *PNAS* and *Global Change Biology*
 - Co-led development of >\$1M in competitive proposals for new forest resilience research
- Graduate Research Assistant** 2016 - 2022
Department of Ecology, Evolution, & Organismal Biology, Brown University
- Developed multi-scale assessments of pine species' climate vulnerability using ecological niche models, tree-ring data, and field surveys
 - Published findings on ecological forecasting of climate vulnerability in *PNAS*, *Ecography*, *Nature Climate Change*, and *Global Ecology & Biogeography*
 - Led extensive field campaigns internationally and across the western United States
- Spatial Analyst, University of Montana** 2014 – 2016
- Developed and deployed spatiotemporal wetland distribution models at landscape scales for Greater Sage Grouse conservation across the western US
 - Collaborative interagency project resulted in interactive web platform for management planning and decision-making
- Field Research Technician, Yellowstone Wolf Project** 2013 – 2014
- Extensive backcountry fieldwork in support of high-profile wolf and mountain lion research

PEER-REVIEWED PUBLICATIONS

- Perret, D. L.,** D. M. Bell, H. S. J. Zald. 2025. Reducing fire severity and extent bolsters subalpine forest resilience to global change through key demographic pathways. *Global Change Biology*. <https://doi.org/10.1111/gcb.70052>
- Kissling, M. L., M. C. McDevitt, J. Felis, **D. L. Perret**, P. M. Lukacs, *et al.* 2025. Nesting ecology and movements of an ice-associated seabird, the Kittlitz's murrelet (*Brachyramphus brevirostris*) at the northern edge of its range. *Wildlife Society Bulletin*.
- Perret, D. L.,** M. E. K. Evans, D. F. Sax. 2024. A species' response to spatial climatic variation does not predict its response to climate change. *PNAS*. <https://doi.org/10.1073/pnas.2304404120>
- Perret, D. L.,** D. M. Bell, A. N. Gray, J. D. Shaw, H. S. J. Zald. 2023. Range-wide population assessments for subalpine fir indicate widespread disturbance-driven decline. *Forest Ecology & Management*. <https://doi.org/10.1016/j.foreco.2023.121128>
- Perret, D. L.,** & D. F. Sax. 2021. Evaluating alternative study designs for optimal sampling of species' climatic niches. *Ecography*. <https://doi.org/10.1111/ecog.06014>
- Rosenblad, K., **D. L. Perret**, D. F. Sax. 2019. Niche syndromes reveal climate-driven extinction threat to island endemic conifers. *Nature Climate Change*. <https://doi.org/10.1038/s41558-019-0530-9>
- Perret, D. L.,** A. B. Leslie, D. F. Sax. 2019. Naturalized distributions show that climatic disequilibrium is structured by niche size in pines (*Pinus* L.). *Global Ecology & Biogeography*. <https://doi.org/10.1111/geb.12862>
- Donnelly, P. J., B. W. Allred, **D. L. Perret**, N. L. Silverman, J. D. Tack, V. J. Dreitz, & D. E. Naugle. 2019. Seasonal drought in North America's sagebrush biome structures dynamic ecological minimums for sage-grouse. *Ecology & Evolution*. <https://doi.org/10.1002/ece3.4614>

MANUSCRIPTS IN PREPARATION

- McLaughlin, B. P., M. Kling, S. N. Miller, **D. L. Perret**, B. J. McGill. An ecology-focused, new-user introduction to the Forest Inventory and Analysis (FIA) Database.
- Perret, D. L.,** D. M. Bell, H. S. J. Zald. Plots, planes, pixels: multi-scale data integration for dynamic monitoring of forest mortality and risk from insects, diseases, and drought.
- Perret, D. L.,** D. M. Bell, H. S. H. Zald. Dendroecological indicators of sugar pine (*Pinus lambertiana*) vulnerability and decline across its geographic and climatic distribution.

GRANTS AND AWARDS

- Plots, planes, & pixels: multiscale data integration for dynamic monitoring of forest mortality and risk from insects, diseases, and drought. (submitted 2025)* (\$824,622)
Science PI – **Daniel Perret**, PI – Harold Zald, Co-I – David Bell
NASA ROSES-24 A.60 Ecological Conservation grant proposal supporting data integration and fusion to support mapping and estimation of forest mortality and future vulnerability to biological disturbances.
- Plots, planes, & pixels: biological disturbance agent estimation. (2023)* \$114,000
PI – Harold Zald, Co-PIs – David Bell, **Daniel Perret**
ORISE Fellowship supported by interagency agreement between the US Forest Service Pacific Northwest Research Station and the US Department of Energy.
- Vulnerability assessment of sugar pine, an iconic species in decline. (2022)* \$100,000
PI – Harold Zald, Co-PIs – David Bell, Andrew Gray, **Daniel Perret**
ORISE Fellowship supported by interagency agreement between the US Forest Service Pacific Northwest Research Station and the US Department of Energy.

<i>Does ecological niche theory predict tree performance in exotic communities and climates?</i> (2020)	\$9,935
Doctoral Dissertation Enhancement Grant, The Bushnell Graduate Research & Education Fund.	
<i>Reading between the rings: the climate future of North American pines.</i> (2019)	\$4,596
Graduate Research, Travel & Training Grant, Institute at Brown for Environment and Society.	
<i>Assessing climate-driven extinction risk by measuring performance of North American pines.</i> (2018)	\$4,000
Lewis and Clark Fund for Exploration and Field Research, The American Philosophical Society.	
<i>Assessing climate-driven extinction risk by measuring the population performance of pines.</i> (2018)	\$4,977
Graduate Research, Travel & Training Grant, Institute at Brown for Environment and Society.	
<i>Spatial distribution and morphological variation in the mountain beaver (<i>Aplodontia rufa</i>).</i> (2011)	\$2,600
Stanford University Undergraduate Advising and Research Small Grant.	

ORGANIZED SYMPOSIA

- Demographic perspectives on forest responses to global change.* Lead organizer for organized oral session at the US Forest Service Forest Inventory and Analysis Science Symposium. November 2024.
- Forest Inventory Data for Global Change Ecology: Opportunities, Challenges, and Innovations.* Lead organizer for organized oral session at Ecological Society of America. Portland, OR. August 2023.
- An Introduction to Estimating Ecological Change with the US Forest Service Forest Inventory and Analysis (FIA) Database.* Lead organizer for workshop at Ecological Society of America. Portland, OR. August 2023.

TALKS AND POSTERS

- Forest health and climate resilience in the Blackfoot Watershed.* Aspen Ecology Workshop, Greenough MT. July 2025 (planned).
- Plots, planes, and pixels: comparing and integrating datasets to improve forest mortality detection.* US Forest Service FIA Science Symposium. November 2024.
- Reducing fire severity and extent bolsters subalpine forest resilience to global change through key demographic pathways.* US Forest Service FIA Science Symposium. November 2024.
- Changing world, changing forests: global, continental, and regional perspectives.* Invited seminar, University of Montana. Missoula, MT. April 2024.
- Range-wide population assessments for subalpine fir: widespread disturbance-driven decline.* USFS Pacific Northwest FIA Operational Meeting. January 2024.
- Interactions between climate and disturbance are driving widespread subalpine fir decline.* FIA Stakeholder Science Meeting. November 2022.
- Interactions between climate and disturbance are driving widespread subalpine fir decline.* MtnClimate. Crested Butte, CO. September 2022.
- Population-specific responses determine species' vulnerability to climate change: a spatiotemporal case study of ponderosa pine.* Ecological Society of America. Montreal, Canada. August 2022.
- Tree ring records reveal that climate envelope approaches underestimate population vulnerability to climate change.* International Biogeography Society. Vancouver, Canada. January 2022.
- Niches, tree-rings, and invasive forests.* Brown University EEOB Departmental Seminar. Dec 2020.
- Tree rings and niche models as tools for predicting forest responses to climate change.* Earth, Itself. Brown University. March 2019. (poster)
- Is all niche space created equal? A framework for exploring the structure of the niche.* International Biogeography Society. Malaga, Spain. Jan 2019. (poster)
- Climatic disequilibrium is structured by niche size.* International Biogeography Society, Climate Change Biogeography. Evora, Portugal. March 2018.
- Niche size structures climatic disequilibrium in pines (*Pinus L.*).* New York Species Distribution Modeling Symposium. Yale University. December 2017.

TEACHING

Conservation Biology (Biol 1470) , teaching assistant <i>Brown University</i>	2020
Guest lectures: <i>Wildlife Management; Wildlife, Climate, and Conservation</i>	
Conservation Biology (Biol 1470) , teaching assistant <i>Brown University</i>	2017

PROFESSIONAL WORKSHOPS

Dendrochronology Intensive Summer Course Laboratory of Tree-Ring Research, University of Arizona	2018
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PROFESSIONAL SERVICE

Reviewer for *PNAS*, *Science Advances*, *New Phytologist*, *Ecography*, *Global Ecology & Biogeography*, *Ecosphere*, *Ecology & Evolution*, *Forest Ecology & Management*

REFERENCES

Harold Zald

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